

# Marllon Rodrigues

## Software Engineer

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### Technical Skills

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**Languages & Frameworks:** C#, ASP.NET Core, Entity Framework Core, SQL Server (T-SQL, SSMS, SSRS), Python, TypeScript, Node.js (Fastify, Express), React, Vue.js.

**Architecture & Practices:** Microservices, DDD, SOLID, REST API design, multi-tenant SaaS, RBAC & security hardening, predictive analytics (Z-score, ETS).

**Infrastructure:** PostgreSQL, Prisma, Redis, Docker, GitHub Actions CI/CD, Cloudflare R2, industrial IT/OT integration (SCADA, Modbus, IEC 61850).

### Experience

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**Automation Engineer, Automovaa (Singapore)** Jul 2026 – Present

- **AMHS System (MAS):** PLC programming for an Automated Material Handling System (pallet AGV/stacker) using Schneider Electric EcoStruxure Machine Expert; safety logic, subsystem integration (BIN, PCon, Vault Lane), and UAT2 validation.
- **Safety Fix & Codebase Migration:** Identified and resolved a critical E-Stop safety defect across multiple subsystems; led full audit-driven port-over of PB1 logic to PB2, resolving divergences across the entire codebase.
- **Internal Documentation Tool:** Built a React web app (MAS-web) consolidating project knowledge — IO lists, subsystem diagrams, changelog — used by the engineering team for onboarding and daily reference.

**Full Stack Engineer / Tech Lead, Bunkerchain (Singapore/Remote)** Jun 2025 – Sep 2026

- **Multi-tenant SaaS Platform:** Built a B2B platform from scratch with full multi-tenancy, granular RBAC, Kanban, SSE real-time notifications, and file uploads via Cloudflare R2. Stack: Fastify v5, Prisma, PostgreSQL, Redis.
- **Security & Authorization:** Identified and closed 10+ cross-tenant authorization bypass vulnerabilities across 12+ user roles; step-up authentication on sensitive operations; hash-chained audit logs with global sequence integrity.
- **Cryptographic Verification System (VPS):** Trust verification for physical fuel sample bottle seals using ECDSA digital signatures, a public key registry, seal lifecycle state machine, and 6 REST API endpoints.
- **Maritime Platform (MVP 3.0):** FSM-driven port call lifecycle with regulatory holds and multi-signatory workflows; full-stack delivery from architecture to live client demo approval.

**Automation Analyst / Field Engineering Tools, Tyconnex** Feb 2026 – Jul 2026

- **Field Engineering Tool:** Built a custom offline web app consolidating 3,432 equipment records, Modbus address tables, and IP mapping for 1,339 devices; used daily by the field team on two FPSO platforms (Petrobras P80/P83).
- **RDO Generator:** Automated daily report pipeline (Python + PyYAML + LaTeX + Docker); 27+ professional PDF reports documenting 231 hours of on-site commissioning.

**Systems Developer, MAIMA Technology** Aug 2024 – May 2025

- **Predictive Maintenance System (C# / ASP.NET Core):** Designed and built a distributed system for machine downtime prediction based on product quality signals; microservices architecture with Image Analysis, Data, and BFF layers; anticipates critical failures before occurrence and triggers automated alerts to operations.
- **Domain-Driven Design & REST APIs:** Applied DDD and SOLID principles across service boundaries; implemented REST APIs in ASP.NET Core consumed by multiple internal clients; managed SQL Server databases with raw T-SQL for complex reporting queries.
- **AI-Powered CI/CD & Anomaly Detection:** Integrated automated code review and security scanning on every commit; applied Z-score and ETS statistical models for anomaly detection in operational data streams.
- **SAP Automation (Python / RPA):** Built Python robots automating 6 critical daily SAP processes, reducing execution time from 8 hours to 2 minutes.

**Automation Technician, S.Build Automation** Jun 2022 – Aug 2024

- **SQL & Data Automation:** Python scripts for mass SQL updates in AVEVA MES (SQL Server); automated tasks across thousands of records and SSRS reporting pipelines.
- **Legacy Migration & Retrofit:** Migration (1995 → 2023) with reverse-engineering of undocumented machines; upgrade of 20+ servers via ThinManager; on-site commissioning in critical industrial environments.

Certifications

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- **.NET Developer Bootcamp** (DIO – 93h)    |    **Java OOP** (Udemy – 54.5h)    |    **Node.js Fundamentals** (DIO – 30h)
- **CSI AVEVA** Manufacturing Execution System (MES) 2020 – Operations & Performance Exams
- **CSI AVEVA** Application Server 2023 Exam

Education

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| UNINTER, Systems Analysis and Development            | 2023 – 2025 |
| SENAI Roberto Simonsen, Computer Networks Technician | 2021 – 2022 |

Languages

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- English (C1 – Advanced)
- Portuguese (Native)